Practical 4

Aim: Implementation of K-Means Clustering

Source-Code:

```
#Necessary Libraries
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.datasets import make_blobs

#Generate Data
X, _ = make_blobs(n_samples = 300, centers = 4 , cluster_std = 0.60,random_state = 0)

#Plot the data points
plt.scatter(X[:,0],X[:,1])
plt.title("Data Points")
plt.show()

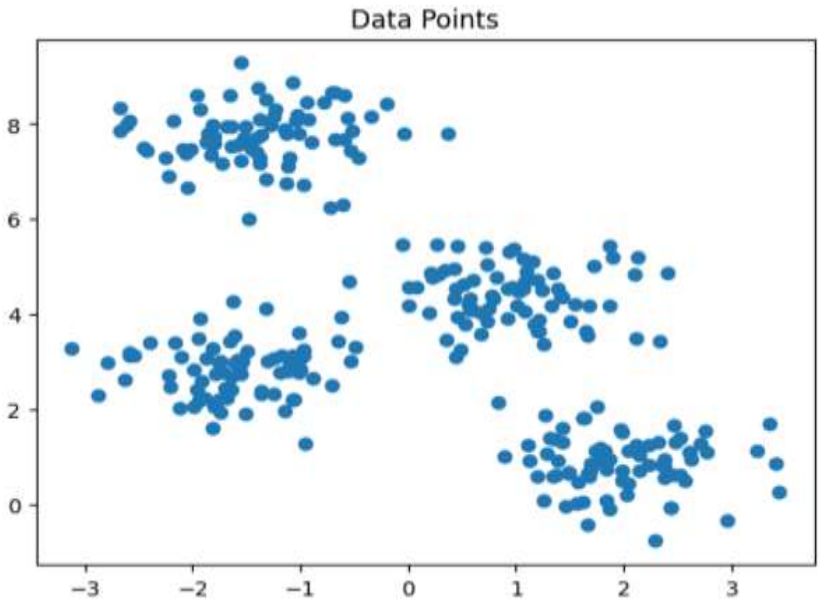
#KMeans clustering
kmeans = KMeans(n_clusters = 4)
kmeans.fit(X)

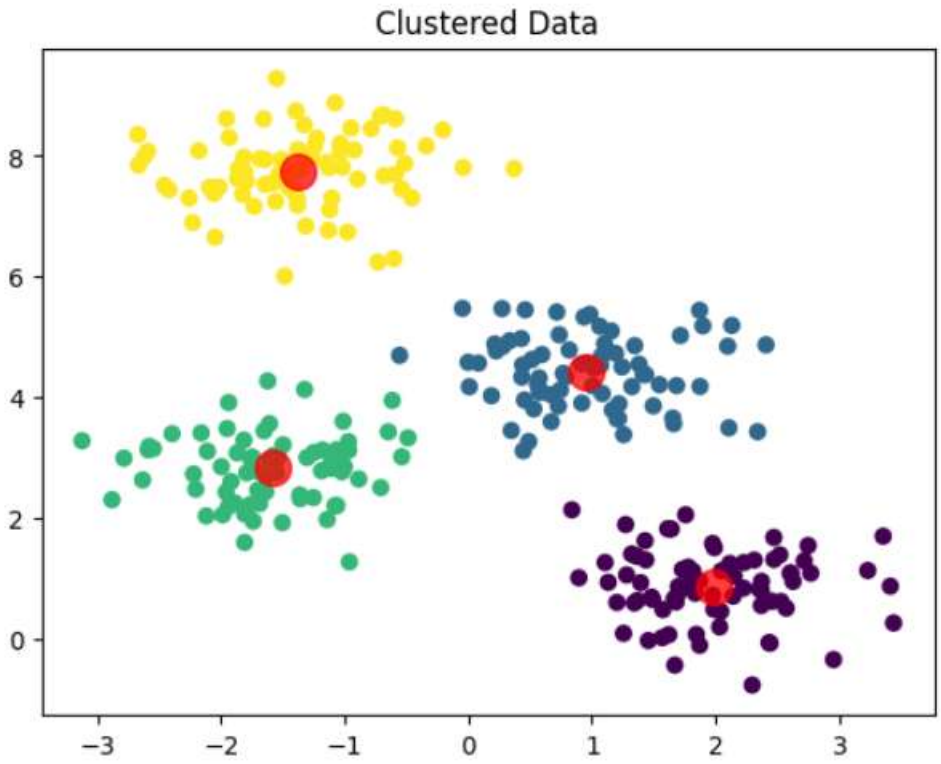
#Getting the cluster centers and labels
centers = kmeans.cluster_centers_ #returns the coordinates of the

#cluster center found by the algorithm
labels = kmeans.labels_ #Returns the labels for each data point , indicating which cluster it
has

#been assigned to.
#Plot the clustered data
plt.scatter(X[:,0],X[:,1],c = labels)
plt.scatter(centers[:,0],centers[:,1],alpha = 0.75,s=200,color = 'red')
plt.title("Clustered Data")
plt.show()
```

Output:




Clustered Data
8
6
4
2
0
−3
−2
−1
0
1
2
3